

Ontology management in enterprises

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In today's increasingly competitive business market, many large organisations have begun to research and develop flexible enterprises in order to be able to quickly respond to market opportunities. They are seeking ways to maximise the power of information assets stored across hundreds of their databases and application programs by bringing them into open interoperable environments. However, this effort has been seriously hindered by various kinds of heterogeneity. Each system has its own domain model for its environment and efficient task operations. However, in order for systems to communicate and co-operate effectively, a shared domain model is required. The semantics of the common, and each local, model must be captured explicitly and formally to enable meaningful information exchange.

The semantics of diverse information resources are captured by ontologies — definitions of terms as used in data sources, i.e. concepts and the relationships between concepts. When defining the relationships between data sources, we rely on ontologies to make the meaning of the different vocabularies used explicit. This paper explores the role of ontologies in enterprises, and proposes a methodology for managing enterprise ontology resources and a suite of support tools.

1. Introduction

Information systems play an expanding role in all aspects of business. Most large organisations now regard information systems as critical to their commercial success. With the advent of the Internet and electronic commerce, companies are now seeking ways to maximise the power of their information assets stored across hundreds of databases and application programs by bringing them into open interoperable environments. These systems embody essential company information, represent large investments, and have become very reliable over the years. However, they seriously hinder the responsiveness of an organisation. Invariably there is a need to integrate information systems and share information in ways not considered when those systems were first deployed. For example, in the context of e-Business there is a need to dynamically and openly interoperate with other enterprises in support of supply chains. Currently it is difficult to exchange meaningful information about products, services and customers between the information systems of different enterprises.

The main cause of this difficulty is heterogeneity [1, 2]. This includes different hardware, operating systems, data models, data semantics and data quality. Often, data that is available is subject to implicit assumptions which mean that it cannot be reused out of context. Furthermore, data is often duplicated and labelled under different terms. These problems are increasing as companies go global through mergers and acquisitions necessitating the integration of different enterprise information systems. Similarly,

electronic commerce and the automated linking of supply chains require seamless access to information across disparate information sources.

Many technologies have been developed to tackle the interoperability among heterogeneous systems. Middleware, such as CORBA and DCOM, has been successfully deployed to allow interoperation among different platforms and software. XML and associated technologies such as DTD and RDF are perceived to be key enablers for the exchange of information over the Internet and corporate intranets by overcoming the syntax and data model heterogeneity. However, XML without agreed-upon DTDs does nothing to support integration at the semantic level. Many attempts have been made to set up standards for information exchange. However, they are often too rigid to reflect change and local needs. Alternative approaches need to be investigated. The semantics of information resources can be captured by ontologies, which provide formal definitions of the terms used in data sources. When we want to identify the relationships between the different terminologies of data sources, we can use ontologies to make the implicit meaning of their different terminologies explicit. It is then possible to dynamically locate relevant data sources based on their content and to integrate them as need arises.

It is intrinsically difficult to develop a machine-processable and understandable vocabulary because hidden

contextual information needs to be made explicit. When an application is built and run within a single context, there is no need to make implicit assumptions explicit. However, when we want these applications to run in an open environment, implicit assumptions have to be made explicit to avoid misunderstanding. We believe ontology engineering will be a major part of any future application development. Thus, support environments and tools are required to assist ontology designers with defining, updating, reusing and mapping terms between ontologies. This paper explores the role of ontologies in enterprises and proposes a methodology for managing enterprise ontology resources and a suite of support tools.

The rest of the paper is structured as follows. Section 2 discusses the role of ontologies in an enterprise. Section 3 introduces ontologies and their representations. Section 4 discusses information integration and a new information infrastructure. Section 5 describes ontology management life-cycle methodologies and proposes a new architecture, a new methodology and support tools. Then we briefly discuss the related work and conclude the paper with future directions.

2. Ontology in enterprises

Enterprises rely on information systems for their operations. When an enterprise seeks to provide service-oriented businesses, their existing information systems need to be given additional flexibility and integrated according to service requirements. However, these so-called legacy systems are often independently developed databases that support often isolated or stove-piped business applications. The reuse of these systems to support new business applications is a major challenge to all enterprises.

Heterogeneity is the main difficulty. There are various kinds of heterogeneity. Amit Sheth [3] classifies heterogeneity into four categories: system, syntax, structure and semantic —

- system heterogeneity includes incompatible hardware and operating systems,
- syntax heterogeneity refers to different languages and data representations,
- structure heterogeneity includes different data models such as relational and object-oriented models,
- semantic heterogeneity refers to the meaning of terms.

Examples of semantic heterogeneity are that people in different places often use different terms to refer to the same or similar products. Worse still, the same term may be used to refer to different products.

An ontology for a domain is a conceptualisation of the world in terms of the objects, qualities, distinctions and relationships, etc, in that domain. A shared or common ontology refers to an explicit specification of concepts. Such a specification should be an objective (i.e. interpretable outside the systems) description of the concepts and relationships that the participants use to interact with each other, with other programs such as legacy business applications, and with humans. We see that ontologies provide the foundation to build collaborative information systems that meet the information requirements of flexible enterprises.

- Agent communications

A shared ontology enables agents to have meaningful communications. Although the participating agents for collaborative business services [4] may have a common shared domain, they often have different views of the same domain and may have adopted different models best suited to their local services. When the vocabulary of an agent is captured in an ontology, messages it sends can be interpreted correctly by the receiving agents.

- Horizontal information integration

The number of the underlying information systems involved is potentially large. Invariably only part of the information held by some system needs to be shared with others for any particular service. However, which parts are to be shared and integrated and for what purposes cannot be determined beforehand. A common ontology enables information systems to be integrated horizontally and dynamically.

- Explicit contents and capability description

Agent services and information support system contents must be formally described. These descriptions must be accessible and meaningful to all participating agents. Ontology-based contents and agent-capability descriptions are machine-processable and thus can be correctly interpreted by agents.

- Different levels of abstraction

An ontology is a formal description of the domain, independent of implementation. It includes terms (i.e. concepts) of various levels of abstraction of a given domain.

- System integration

An ontology provides the linkage between the domain model and information system models. The composition of an ontology will eventually pave the way to automatically integrating the underlying information systems.

An ontology also provides a way to enable system designers, system developers, users and managers to have shared semantics of terms, thus reducing misunderstanding.

Ontologies form the basis of the resolution of the heterogeneity of diverse resources, allowing them access to shared information when required. This provides the flexibility and autonomy of agents that are the trademark of flexible or virtual enterprises. Agents can develop their services without concern for how other agents are developed and what information systems they use.

3. Ontology and ontology representation

Ontologies are considered to be a vehicle to make the meaning of terms explicit so that information can be shared with minimal misunderstanding. In applications developed by closely-knit teams, the intended meaning of a term is often implicit, relying on developers' mutual agreement. In an open environment mutual agreement is hard to come by if not impossible. Thus it is crucial that the domain model and vocabulary are represented in a way that enables programs to derive the intended meaning of the terms with minimum human intervention during their execution. This is the task undertaken by ontology research.

3.1 What is ontology?

Although the ontology community seems keen to agree on the use and on the meaning of the term ontology, there is no accepted formal definition for it.

Gruber [5] gave the most quoted definition. He considers an ontology as 'an explicit specification of a conceptualisation'. Here a conceptualisation is an abstract, simplified view of the world that we wish to represent for some purpose. This includes objects, concepts and their relationships. The explicit specification means some declarative representation.

One of the many criticisms to the use of Gruber's definition of ontology is that Gruber seems to make no distinctions between the domain and the states of affair (situations) naming as domain space the structure composed by both of them. For example, consider a domain with the concept *above* (x, y) – x being above y , and a set of objects $\{a, b, c\}$. The situation of **above** (a, b) and **above** (b, c) would be a conceptualisation, and the situation of **above** (b, c) would be another. According to Guarino, Gruber's definition can be rewritten as follows:

'An ontology is a logical theory accounting for the intended meaning of a formal vocabulary, i.e. its ontological commitment to a particular conceptualisation of the world'.

In this view ontology is language dependent while a conceptualisation is language independent.

Gruber's and Guarino's are not the only definitions of ontologies presented in the literature, although they are the most used. Each research group working in the field tends to have its own definition of ontology. However, despite some minor differences, all the definitions refer to the ontology as a common understanding of a domain, which implies that it is a repository of vocabulary for domain knowledge. The vocabulary can contain both formal and informal definitions. In this paper a pragmatic view is taken. We consider ontology as an explicit specification of terms and their relationships where terms are characterised by attributes. Section 5.2 describes in detail the view of ontologies in this paper.

3.2 Ontology representation

Logic has been used in most of the ontology languages proposed. Ontology in logic is a set of sentences where each sentence has an unambiguous meaning and sentences can be combined together via a set of logical operators. Unfortunately, logic is not always computationally appealing nor easy to read. For example, a small sized ontology, with less than 100 entities, with a small set of properties and constraints between entities, can generate thousands of sentences. This will create problems in checking the consistency of the knowledge base and in proving consequences of a set of sentences.

For this reason most languages developed to represent ontologies have some additional formalisms to represent structures. Many languages have been used to represent ontologies at different levels of formalism, ranging from highly informal languages, such as natural language to rigorously formal ones [7].

Ontolingua [5, 8] has been developed as a tool for knowledge sharing and reuse. Ontolingua is based on knowledge interchange format (KIF) [9] and aims to facilitate reuse and interoperability, playing the role of an interchange language. It provides several built-in ontologies that can be incorporated into the one that is to be written, but it has no inference engine. CLASSIC [10] is a knowledge representation system that was not specifically designed for knowledge sharing but that has been recently used also to represent ontologies. CLASSIC is based on description logic (DL) [10], and possesses most of features of semantic networks. CLASSIC provides a reasoning that allows the automatic classification of a new concept in an ontology and can detect inconsistencies.

4. Use of ontologies for information integration

Information integration has been studied for many years in the database community. Recent advances in technology, such as the Internet, has enabled users to access huge amounts of information from different places all over the world. Stimulated by the growth of the Internet there is a growing demand for an understanding of how to integrate multiple and heterogeneous information sources. Research

on the integration of heterogeneous sources includes attempts to recognise and combine relevant knowledge so as to provide a richer understanding of a particular domain. Integration is particularly valuable if it enables communication between different sources while allowing them to maintain their autonomy.

This section reviews different approaches to the use of ontologies in the context of information integration.

4.1 *Different approaches to integration*

The literature concerning the integration of heterogeneous knowledge sources provides different approaches to the problem (see Tamma and Visser [11] for a comparison of the approaches). There are several approaches to the integration of heterogeneous resources ranging from direct translation to the use of structured multiple-shared ontologies. The former is the so-called 'one-to-one' approach, where for each ontology a set of translating functions is provided to allow mappings with the other ontologies. Indeed, this approach requires in the worst case ($n^2 - n$) mapping functions, if there are n ontologies involved. Therefore, using direct translation seems feasible only if there are a few ontologies. An example of such an approach is the OBSERVER system [12] where different ontologies are integrated by defining mappings between a term in a source ontology and its synonym in a destination ontology. If it is not possible to find a synonym the concept is mapped into a hyponym or a hypernym depending on which of the two translations results in the least information loss.

Many architectures comprise a single shared ontology. Whether such an approach is conceptually realistic is a matter of debate [13]. The drawbacks of dealing with a single shared ontology are similar to those of any standard (see also Visser and Tamma [14]). Often, standards are not very convenient to use since they have to be suitable for all potential uses. Also, the task of defining such standards is often lengthy and complicated. Moreover, committing to a standard restricts the degree of heterogeneity that may exist between those using the standards, and, last but not least, standards — by their nature — resist change, partly due to the aforementioned reasons. Some integration projects with a single shared ontology are KRAFT [8, 15], SIMS [16], and InfoSleuth [17].

In contrast to an approach in which all resources share one body of knowledge, the architecture of multiple ontologies locates shared knowledge in multiple but smaller shared ontologies. This approach, which is thought to be more flexible and scalable, is referred to as ontology-based resource clustering, or in short, ontology clustering [13, 18]. Resources no longer commit to one comprehensive ontology, but are clustered together on the basis of the

similarities they show in the way they conceptualise the common domain.

Multiple ontologies are normally organised into a hierarchy. Concepts in the top level of the ontology hierarchy provide general definitions that are shared by all of the resources, whereas the lower levels of the ontology hierarchy provide definitions of concepts that extend and characterise the concepts defined in the higher levels. These lower-level ontologies make explicit the conceptualisation of one or more resources. The co-existence of heterogeneous sibling ontologies is accomplished by the sibling ontologies providing definitions that extend the definitions in the parent ontology in different ways.

4.2 *Mediation-based information integration*

Most of the new approaches to information integration, such as the KRAFT project, are based on the mediation architecture proposed by Wiederhold [19, 20]. Mediators provide intelligent middleware services in information systems by matching resources to applications, where the resources are autonomous and heterogeneous. Services provided by mediators are intended to provide the customer with added value, and this added value has to be such that the drawbacks of introducing a further level in the architecture are at least balanced. Part of the offered services are those providing knowledge that permits interoperation between the heterogeneous underlying information resources through ontology translation.

To match resources with applications, mediators identify the most relevant resource for the user's needs and retrieve the relevant data, translating it to a common representation and ontology in order to provide an answer to the user's query. In such a task mediators can be supported by facilitators, which are responsible for identifying the resources and data conversion.

The general architecture of a mediation-based information system is shown in Fig 1.

4.3 *Use of ontology for agent communication*

A framework that allows agents to deal with explicit, declaratively represented ontologies is described in the specification of the FIPA Ontology Service [21]. FIPA distinguishes explicit and implicit ontologies. Implicit ontologies are encoded with the software implementation of the agents themselves and thus are not formally published to an ontology service. Ontologies that are only written in a book and cannot thus be managed and queried by a piece of software are regarded as implicit. Ontologies play a major role in the FIPA model of communication — what is communicated is determined by the sender's intention, the receiver's recognition of it, and the assumption that the two agents share a common ontology for the domain of

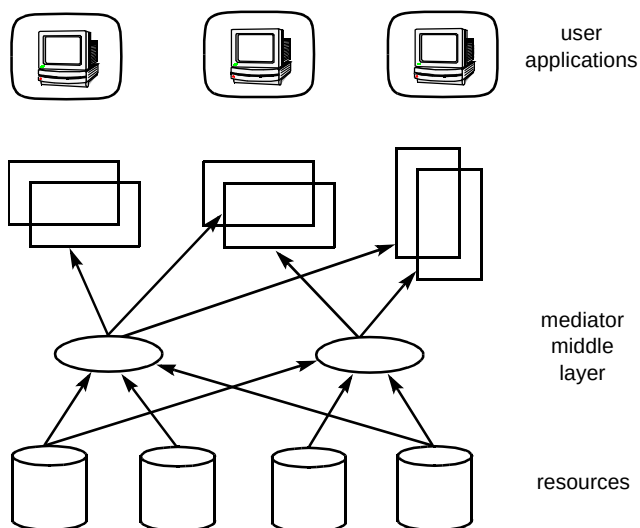


Fig 1 Mediator-based integration.

discourse. The definition of the communicative intentions is covered by the specifications of the FIPA Agent Communication Language [22] that provides a catalogue of communicative acts with their explicit semantics, and a catalogue of interaction protocols based on these communicative acts. Figure 2 shows the FIPA reference model for a possible infrastructure of the organisation of information in a virtual enterprise.

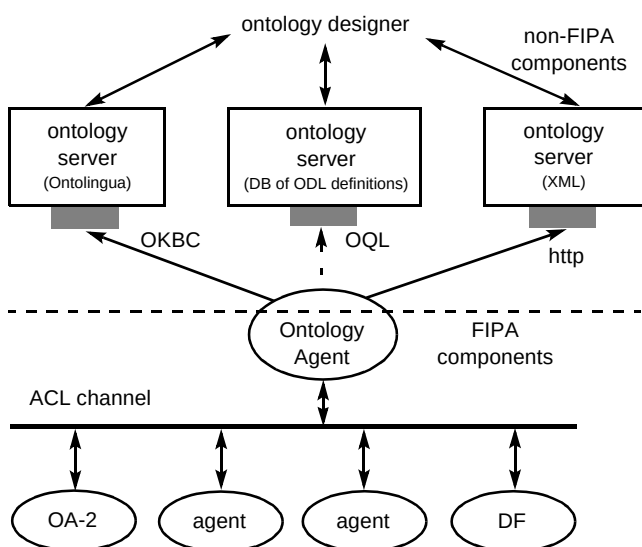


Fig 2 Reference model of the FIPA ontology service.

Ontologies are stored at an ontology server in the upper part of Fig 2. In general, several such servers and ontology repositories exist with different representation formalisms, different interfaces, and different capabilities. The value of the existing legacy systems and the dynamics of the enterprise (processes and affiliated companies may be added, modified, or replaced over the time) discourage the practice of making all the system homogeneous while encouraging the implementation of systems able to deal with heterogeneity.

In the lower part of Fig 2, there are the agents that need to co-ordinate and co-operate to achieve their goals and implement the enterprise processes. The ontology agent (OA) has been envisioned by FIPA as the component that mediates between other agents and ontology servers. It provides access to their services in a unique way that is well integrated with the FIPA Agent Communication mechanism.

An agent that needs to converse in a given ontology may query the OA to discover the existence of the ontology and the relationships with other ontologies. In some cases, an agent may have a need to translate between two ontologies, because, for instance, a legacy database calls a field a different name to that used by another data source. The OA might also be used to provide such a translation service to the agent community, both translating expressions between different ontology representation languages and between different ontologies themselves. An agent may also use the services of the OA in order to access the ontology, for instance to update the ontology, or to query relationships between symbols, or for the definition of symbols (e.g. what are the slots and facets associated with a frame). This feature might be used, for instance, at run time to ask the user for information about an entity or, conversely, to explain the meaning of a term to the user via a conceptual graph (e.g. a candle is a 'part-of' an engine that is a 'part-of' a car, and the number of candles must always be even).

The FIPA specification defines the ontology, the communicative acts, and the interaction protocols (in short, the interface) that should be used by agents to request the services of an ontology agent. In addition to that, because one of the goals of this specification is to allow agents to talk about and manipulate explicit knowledge, it defines a standard meta-ontology and knowledge model. It describes the primitives used by a knowledge representation language, like concepts, attributes and relations. FIPA has adopted the knowledge model of OKBC (open knowledge base connectivity) [23] that supports an object-oriented representation of knowledge with classes, slots and facets. This choice must not be confused with the language to represent ontologies. This has not been mandated by FIPA (even if it is highly desirable) — the OKBC knowledge model and the meta-ontology serve the purpose of the interlingua for knowledge that is being communicated between agents.

5. Ontology management in DOME

The use of databases has produced vast amounts of electronic information. The advent of the Internet has the potential to make this information accessible from almost anywhere, at anytime, by anyone. This situation can result in information poverty as well as information wealth. Although the information required to solve some problem might exist somewhere in a network, it is not easy to locate

the relevant data. Once found, the data can be of varying quality and credibility, making it difficult to interpret. The cost of use and reuse is high.

In order to benefit fully from the networked information it is vital for organisations to have a one-stop knowledge shop for quality information with certainty, authority and consistency. It acts as a corporate knowledge repository that will support the recording of the semantics of implicit or tacit knowledge in information systems and how they are linked to company-wide information, such as data standards, data models, metadata, business models, business logic and business policy. The knowledge shop should also support conflict resolution, translation and knowledge level communications through abstraction, aggregations, and so on.

DOMÉ — a domain ontology management environment — is a research and development project investigating techniques to utilise ontologies in building corporate knowledge shops. Information in the knowledge shop is encoded using ontologies that define machine-processable term semantics. The purpose of DOMÉ is to produce a data semantics management environment, which includes a principled methodology and a suite of tools to support the ontology development lifecycle.

5.1 DOMÉ architecture

DOMÉ is designed to be used by knowledge shop administrators as well as knowledge users. For knowledge shop administrators DOMÉ provides a set of tools for

bringing data sources to the knowledge shop. For knowledge shop users DOMÉ provides a set of tools and APIs to browse, query and retrieve data from the data sources registered with the knowledge shop. Figure 3 shows the DOMÉ architecture. It consists of four main components:

- generic ontology that provides a company-wide shared information view that is often derived from company wide data standards,
- an ontology repository that encodes the contents and semantics of disparate information resources including legacy databases and web-based information sources,
- mapping database that records the mapping information among data items,
- ontology reasoner that maintains the consistency of the knowledge shop.

Ontology administrators interact with the knowledge shop through the tools available to construct ontologies from their data sources. One of these tools is the ontology extraction tool called XRA. It uses the software reverse engineering approach to extract an initial ontology from given data sources and their application programs. The initial ontology contains the entities, relations, functions and instances. XRA [24, 25] can take relational database schemata or object-oriented models and application programs to extract entities, relationships between entities and functions. This initial ontology will be refined by ontology designers. As database schemata are developed

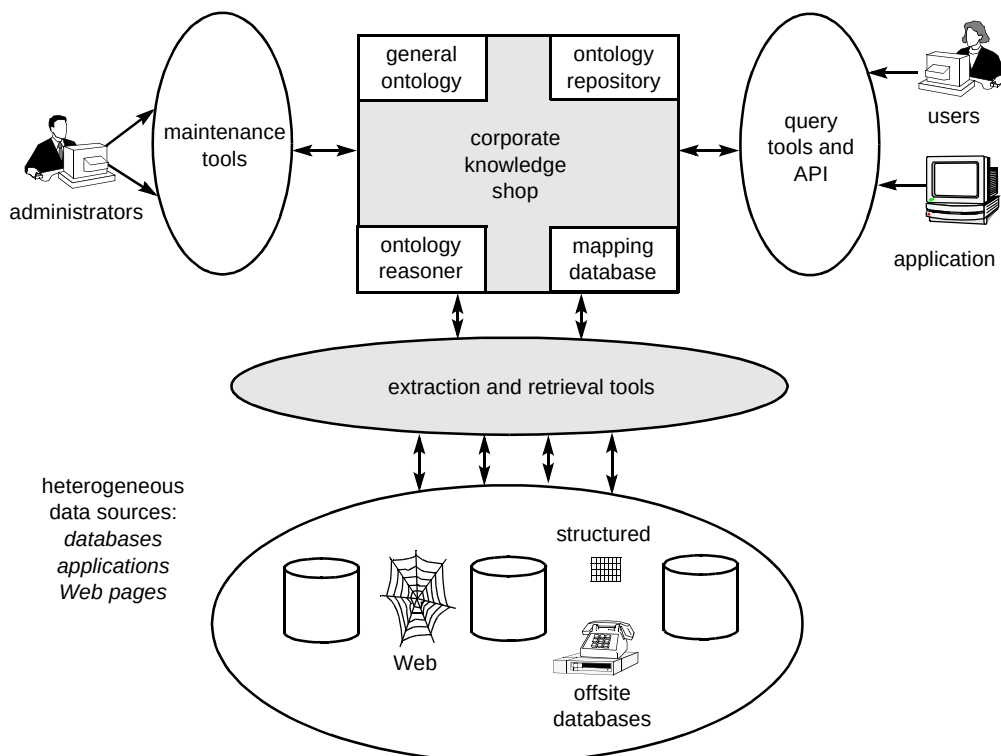


Fig 3 DOMÉ ontology management system components.

with respect to machine-level operations, many semantic relationships can only be discovered from application programs. The concepts discovered from the application programs reflect the correct usage of the data sources that are often not documented.

Ontologies in DOME are implemented in the description logic CLASSIC [17]. In addition to checking ontology consistency, we are developing a rewrite engine on top of CLASSIC to fuse or transform ontologies.

The query tools and APIs that allow users and applications to interact with the ontology repository are based on OKBC [23]. OKBC was developed in a DARPA project and has been adopted by FIPA [21]. DOME is also developing mediators which allow transparent access and retrieval data from the disparate data sources managed by DOME.

5.2 Ontology structure and organisation

In DOME, an ontology consists of terms denoting concepts, their relationships and constraints. A concept is characterised by attributes that are necessary conditions which all objects of the concept must satisfy. This definition is intentionally similar to relational database schema definitions as well as object-oriented database classes because DOME is focusing on constructing ontologies for structured data sources.

Ontologies in DOME incorporate a specialisation hierarchy with lower level terms having close links with ER models and database schemas. Figure 4 shows the relationships of ontology, ER models and database

schemata. Ontology is seen as domain-oriented concepts. It includes abstract concepts and specifies domain-level constraints that can be used for knowledge-level reasoning. Ontology is suited to represent high-level information requirements. Schemata and classes are data-level concepts that are implementation dependent. They are designed to optimise procedural operations. Constraints at this level are operational constraints. Many domain constraints are not explicitly represented at this level. Ontologies can be used to define database schemata and indeed one ontology can be used to define different schemata.

To truly specify a term's meaning is not currently possible. Some assumptions have to be made. We distinguish two classes of ontology terms — primitive terms and defined terms. The semantics of primitives are not specified. When ontology mapping is required, primitives have to be mapped manually. For this purpose we have used a mapping database and attribute semantic clusters to support this task. Once primitives have been mapped, defined terms could be eventually mapped automatically through a rewrite-engine using the mapping database.

5.3 DOME ontology development process

The DOME ontology development process combines the top-down and bottom-up processes [7]. Figure 5 shows the overview of the DOME ontology development process.

The top-down process (from BT standard, WordNet and Cyc of Fig 5) starts with domain analysis to identify the key concepts of the domain. These key concepts can often be obtained by re-engineering data standards that are defined in large organisations. After that competency questions [26]

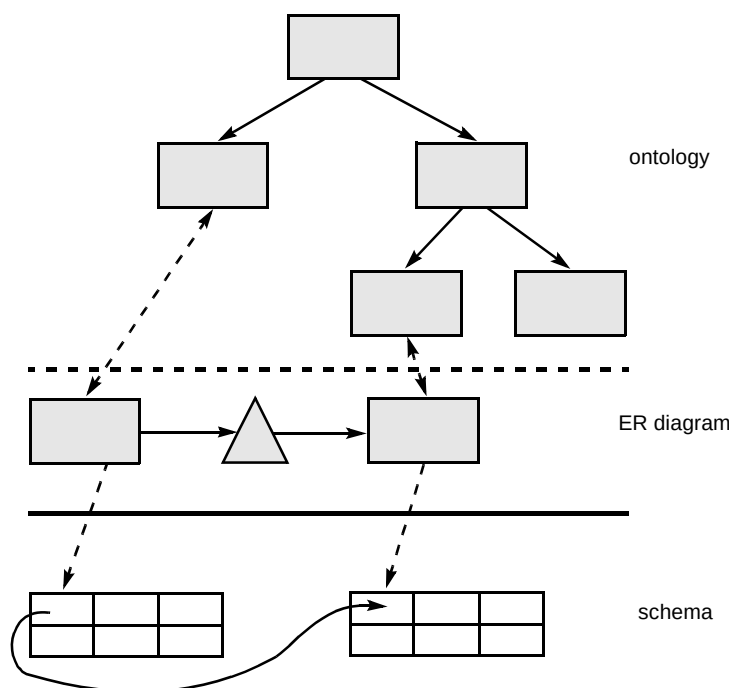


Fig 4 The relationship between ontology and the underlying data sources.

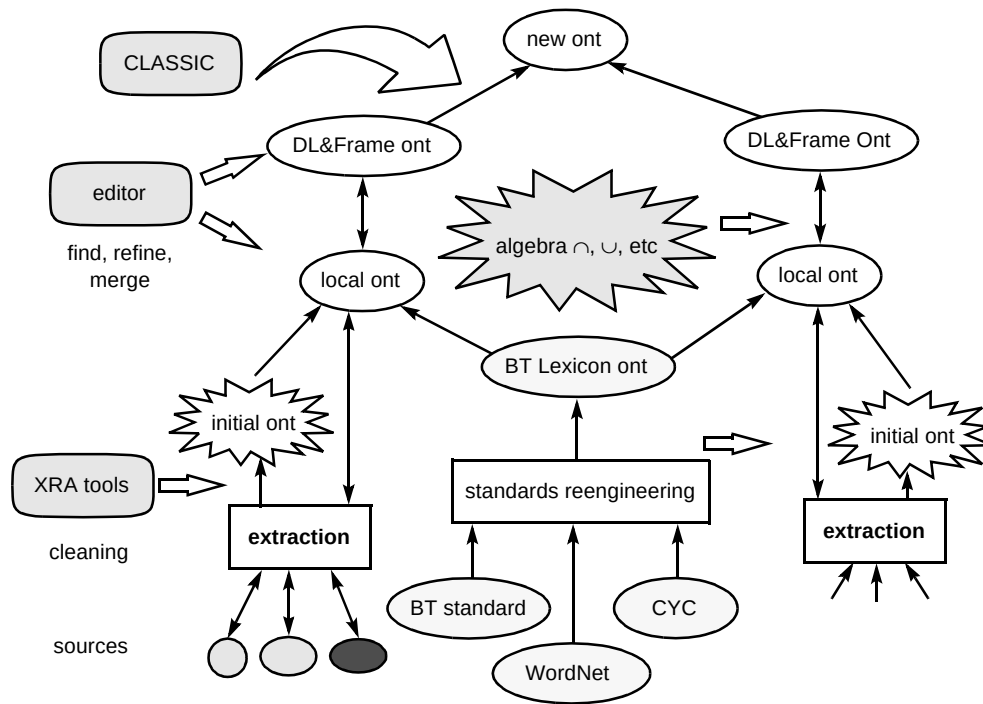


Fig 5 Overview of DOME ontology development process.

are defined, which can be used to guide ontology development and to measure the quality of the ontology obtained. This process produces a general domain ontology (denoted as BT Lexicon Ont in Fig 5). As most of the data standards are flat in structure, lexicon-level ontologies such as WordNet [27] and CYC [28] will be required to organise and turn data standards into hierarchical ontologies. DOME is building tools to assist the use of WordNet, CYC, etc.

The bottom-up process (from extraction of Fig 5) starts with the underlying data sources that need to be re-engineered. We are developing tools in XRA to use software re-engineering techniques to extract ontology from data sources. XRA takes relational database schemata and application programs to produce an initial ontology which includes entities, relations and functions. Application programs often contain new entities and relations that are not present in data schemata.

The task of bridging the bottom-up ontology with the top-level domain ontology is supported through the provision of ontology editing software. The ontologies are implemented in CLASSIC [10], allowing for the consistency of the ontologies to be checked.

6. Related work

Ontology has been studied by many projects both in academia and industry. Most of these projects have provided some level of ontology development support. However, few serious attempts exist to provide a comprehensive ontology building environment which we have argued are crucial for

virtual enterprise development. Among these few efforts we mention Ontolingua [5], SKC [29] and DODDLE [30] although only Ontolingua provides a completely developed environment to build and integrate the ontologies. The other two projects are still in development stage.

Ontolingua [5] of Knowledge Sharing Lab, University of Stanford, is a large project focusing on ontology development. It has built sophisticated tools for developing and maintaining frame-based ontologies. It focuses on formal ontology specifications, reuse, and translation to different ontology implementation systems. It is not addressing problems related to legacy systems and tools to merge ontologies.

SKC [29], also from University of Stanford, is another project focusing on ontology development. SKC is researching ontology composition by developing an ontology algebra. If it succeeds, it would greatly ease ontology development. In DOME, we have used an incremental mapping database to merge ontologies.

Another environment to build ontologies is DOODLE [30], a tool developed at the School of Information of the Shizuoka University. In order to build an ontology the user gives a (unstructured) set of domain terms to DOODLE. These terms are then linked to a machine-readable dictionary (MRD) by the spell match, resulting in a hierarchically structured set of all the nodes on the path from these terms to the root of an MRD. As a matched node (concept) from an MRD sometimes has one or more senses,

the system supports the user in choosing the sense that is most suitable for the user's purposes.

7. Conclusions

With the phenomenal growth of electronically available information and the emergence of electronic commerce, enterprises require collaborative information systems that help with access to information, support decision making and aid in task execution. In this context we have discussed the main obstacle, i.e. semantic heterogeneity, and explored the role of ontology and ontology-based techniques in solving the problem. We then proposed and described an ontology management environment with a methodology and a suite of tools required. We have recognised the issue of defining the meaning of vocabulary as the most critical problem for information sharing and argued that an ontology development environment will reduce system development time and facilitate system interoperability. The ontology in DOME is linked to database schemata and an incremental mapping database is used to record the mapping information. The solution to the semantic heterogeneity problem calls for the adaptation and extension of existing approaches in the areas of database, artificial intelligence and distributed systems technology. DOME is exploring the combination of techniques from those areas.

Acknowledgement

We would like to thank Pepijn Visser and members of IBSR group for contributions to various issues discussed in this paper.

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